WORKED SOLUTION

STEP II 2022, Question 8

Pure • Invariant Lines of Matrices

STEP II 2022, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A line through the origin is invariant when every point on it is sent to a point on the same line, so it is enough to see where one direction vector goes. Writing the line as $y = mx$ and asking that $\mathbf{M}\binom{1}{m}$ be a multiple of $\binom{1}{m}$ turns everything into a quadratic in m. After that, parts (ii) and (iii) are about the sum and product of its roots, and the vertical line, which has no gradient, needs a separate look every time.

Solution

Part (i)

Take the line $y = mx$, with direction $\binom{1}{m}$. Its image is

$$\mathbf{M}\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}a+bm\\c+dm\end{pmatrix}.$$

Because $\mathbf{M}$ is linear, the line is invariant exactly when this image lies on $y = mx$, that is when

$$c + dm = m(a + bm) \quad \Longleftrightarrow \quad bm^2 + (a-d)m - c = 0.$$

So every non-vertical invariant line has a gradient that satisfies this equation, and every root gives an invariant line.

Before calling the gradients "the roots" we must check that this really is a quadratic. The y-axis has direction $\binom{0}{1}$, whose image is $\binom{b}{d}$; this lies on the y-axis exactly when $b = 0$. So if neither invariant line is the y-axis, then $b \neq 0$. The equation is then a genuine quadratic, with at most two roots. Both invariant lines are non-vertical with different gradients, so their gradients are two distinct roots. That means they are exactly the two roots, as required.

If one invariant line is the y-axis, then $b = 0$ and the equation collapses to $(a-d)m - c =$

0. Since $a \neq d$ (this is where that condition is used), the other invariant line has gradient

$$m = \frac{c}{a-d}.$$

Part (ii)

Case $b \neq 0$. Let the gradients be m_1 and m_2. From part (i),

$$m_1 + m_2 = -\frac{a-d}{b}, \qquad m_1 m_2 = -\frac{c}{b}.$$

If the lines make angles α_1 and α_2 with the x-axis, then

$$\tan(\alpha_1 - \alpha_2) = \frac{m_1 - m_2}{1 + m_1 m_2}.$$

An angle of 45° between the lines means this equals ± 1, so $(m_1 - m_2)^2 = (1 + m_1 m_2)^2$. Writing the left side as $(m_1 + m_2)^2 - 4m_1 m_2$ gives

$$\frac{(a-d)^2}{b^2} + \frac{4c}{b} = \left(1 - \frac{c}{b}\right)^2.$$

Multiplying by b^2 gives $(a-d)^2 + 4bc = (b-c)^2$, that is,

$$(a-d)^2 = (b-c)^2 - 4bc.$$

Case $b = 0$. One line is the y-axis and the other has gradient $c/(a-d)$. For it to make 45° with the y-axis it must have gradient ± 1, so $c = \pm(a-d)$ and $(a-d)^2 = c^2$. With $b = 0$ the right-hand side of the required result is also c^2, so the result holds in this case too.

Part (iii)

Ask yourself What does "equal angles with $y = x$" mean in terms of gradients?

Two different lines through the origin make the same angle with $y = x$ exactly when one is the mirror image of the other in $y = x$. Reflecting in $y = x$ swaps the coordinates, so it sends $y = mx$ to $x = my$, which is $y = \frac{1}{m}x$ when $m \neq 0$. It sends the x-axis to the y-axis and the y-axis to the x-axis.

Case $b \neq 0$. Neither line is vertical. If one had gradient 0, its mirror image would be the y-axis, which is not invariant, so the lines would not be symmetric. Otherwise the condition is $m_2 = 1/m_1$, that is, $m_1 m_2 = 1$. By part (i) this is $-c/b = 1$, that is, $b + c = 0$.

Case $b = 0$. One line is the y-axis, so the other must be the x-axis: $c/(a-d) = 0$, that is, $c = 0$. Given $b = 0$, this is again the condition $b + c = 0$.

So in all cases the necessary and sufficient condition is

$$b + c = 0, \quad \text{that is,} \quad c = -b.$$

(When the matrix is symmetric, so $b = c$, the invariant lines are perpendicular instead. Here the off-diagonal entries are opposite.)

Part (iv)

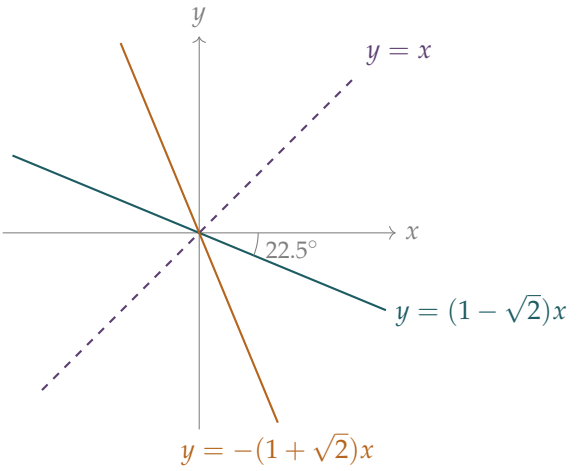
Put $c = -b$ into the condition of part (ii):

$$(a-d)^2 = (2b)^2 + 4b^2 = 8b^2.$$

We need $b \neq 0$: if $b = c = 0$, then $(a-d)^2 = 0$, which contradicts $a \neq d$. Take $b = 1$, $c = -1$ and $a - d = 2\sqrt{2}$, for example $a = \sqrt{2}$, $d = -\sqrt{2}$:

$$\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & -\sqrt{2} \end{pmatrix}.$$

The quadratic of part (i) is $m^2 + 2\sqrt{2}\,m + 1 = 0$, with distinct real roots $m = -\sqrt{2} \pm 1$. So this matrix has exactly two invariant lines, as the question assumes. Their product is 1, so they are mirror images in $y = x$. Their angles with the x-axis are $-22.5°$ and $-67.5°$, since $\tan 22.5° = \sqrt{2} - 1$ and $\tan 67.5° = \sqrt{2} + 1$. They therefore meet at 45°, and each makes 67.5° with $y = x$.



Checks

- For the example in part (iv), $(a-d)^2 = 8$ and $(b-c)^2 - 4bc = 4 + 4 = 8$, so the condition of part (ii) holds. We can also check one line directly: $\mathbf{M}\begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix} = \begin{pmatrix} \sqrt{2}+1-\sqrt{2} \\ -1-\sqrt{2}+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1-\sqrt{2} \end{pmatrix}$, so every point of $y = (1-\sqrt{2})x$ is in fact fixed.
- A quick special case for part (i): for $\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ the invariant lines are the axes. The formula from the y-axis case gives the other gradient as $0/(2-1) = 0$, which is correct.

- Parts (i)–(iii) were confirmed by computer on many thousands of random matrices: the eigenvector directions satisfy the quadratic, pairs at 45° satisfy the identity, and the lines are symmetric about $y = x$ exactly when $b + c = 0$.

Common slips

- Forgetting the vertical line. The quadratic only sees gradients, so the case $b = 0$ has to be handled separately in every part.
- In part (iii), writing $m_1 = -m_2$. That is symmetry about the *axes*. Symmetry about $y = x$ needs $m_1 m_2 = 1$.
- Using $\tan\theta = 1$ without the $\pm$. Squaring $(m_1 - m_2)^2 = (1 + m_1 m_2)^2$ deals with both signs at once.

Beyond STEP

The invariant lines through the origin are the *eigenvector* directions of $\mathbf{M}$, and the quadratic in part (i) is the eigenvector equation $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ with λ eliminated. A real symmetric matrix always has perpendicular eigenvectors (the spectral theorem). This question shows how antisymmetric off-diagonal entries instead make the eigenvectors symmetric about $y = x$. Strang's *Introduction to Linear Algebra* develops eigenvectors and the spectral theorem.

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